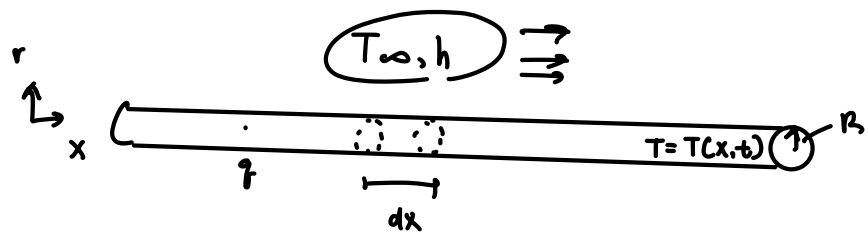
(1) **Given:** $T = T(x,t)$, L (rod length)

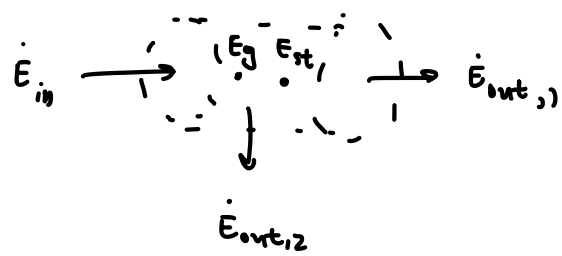
Find: a. energy balance equation for CV; express energy term in the energy balance eq. as function of T

b. derive governing diff. eq. describing temp. profile

c. boundary conditions & initial condition

Schematic





Assumption: R_{rod} very small; T_{rod} along radial direction is uniform; uniform $\dot{q}$; thermal radiation is neglected; h, ρ, and C_p are constant.

Analysis:

a.

$$A = \pi R^2 \Rightarrow dA_s = 2\pi R dx, \quad dV = A dx$$

$$\dot{E}_{in} + \dot{E}_g - \dot{E}_{out} = \dot{E}_{st}$$

$$\dot{E}_{in} = q_x = -hA \frac{\partial T}{\partial x}\Big|_x$$

$$E_{out} = q_{x+dx} + h(2\pi R dx)(T - T_\infty)$$

$$\dot{E}_g = \dot{q} A dx$$

$$\dot{E}_{st} = \rho c_p \frac{dT}{dt} dV = \rho c_p \frac{\partial T}{\partial t} A dx$$

$$q_x + \dot{q} A dx - q_{x+dx} - h(2\pi R dx)(T - T_\infty) = \rho c_p \frac{\partial T}{\partial t} A dx$$

$$\hookrightarrow q_x + \frac{\partial q_x}{\partial x} dx$$

$$\boxed{-hA\frac{dT}{\partial x} + \dot{q} A dx - \left(-hA\frac{\partial T}{\partial x} + \frac{\partial}{\partial x}\left(-hA\frac{\partial T}{\partial x}\right)dx\right) - h(2\pi R dx)(T - T_\infty) = \rho c_p \frac{\partial T}{\partial t} A dx}$$

b.

$$\cancel{-hA\frac{dT}{\partial x}} + \dot{q} A dx - \left(\cancel{-hA\frac{\partial T}{\partial x}} + \frac{\partial}{\partial x}\left(-hA\frac{\partial T}{\partial x}\right)dx\right) - h(2\pi R dx)(T - T_\infty) = \rho c_p \frac{\partial T}{\partial t} A dx$$

$$\left[\dot{q} A dx + kA\frac{\partial^2 T}{\partial x^2} dx - h(2\pi R dx)(T - T_\infty) = \rho c_p \frac{\partial T}{\partial t} A dx\right] \cdot \frac{1}{dx} \cdot \frac{1}{A}$$

$$\boxed{\dot{q} + k\frac{\partial^2 T}{\partial x^2} - \frac{2h}{R}(T - T_\infty) = \rho c_p \frac{\partial T}{\partial x}}$$

$\llcorner \pi R^2$

c. $T(x,0) = T_i$

$T(0,t) = T_1$

$$-h\frac{\partial T}{\partial x}\Big|_{x=L} = h(T(L,t) - T_\infty)$$

(2) **Given:** $L_A = 30\,cm$ $L_B = 10\,cm$ $T_\infty = 20^\circ C$ $T_w = 115^\circ C$

$k_B = 30\ W/mK$ $h = 500\ W/m^2K$ $T_i = 100^\circ C$

Find:

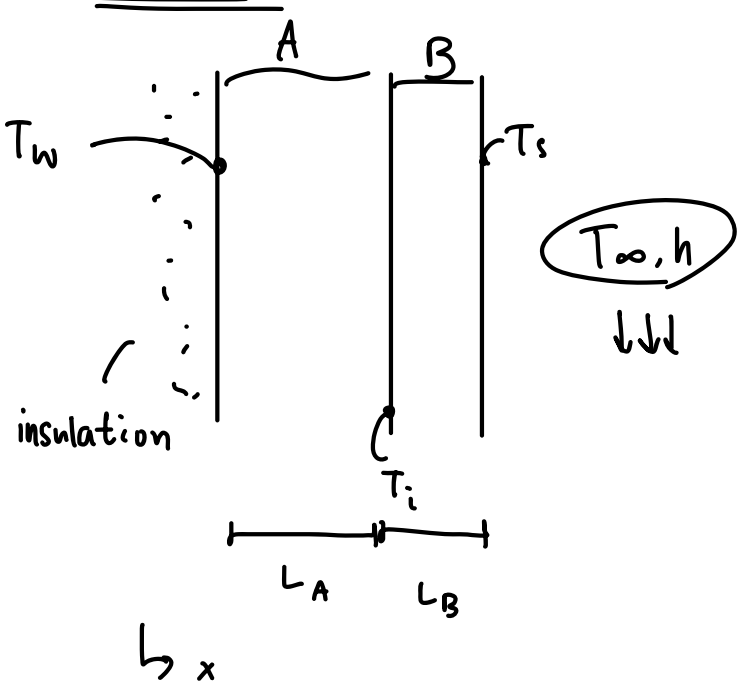
a. right surface temp T_s of wall B

b. uniform volumetric heat generation $\dot{q}$ at steady state

c. thermal conductivity k_A

d. Plot of temp. profile across wall A & B

Schematic



Assumption:
- 1D heat conduction
- uniform $\dot{q}$
- Wall A left is insulated
- no heat generation in wall B

Analysis:

a.
$$q''_{conv} = h(T_s - T_\infty)$$
$$q''_{cond} = \frac{k_B(T_i - T_s)}{L_B}$$

$$h(T_s - T_\infty) = \frac{k_B(T_i - T_s)}{L_B}$$
$$L_B h T_s - L_B h T_\infty = k_B T_i - k_B T_s$$
$$T_s(L_B h + k_B) = k_B T_i + L_B h T_\infty$$
$$T_s = \frac{k_B T_i + L_B h T_\infty}{L_B h + k_B}$$

$$\boxed{T_s = 50^\circ C \ ; \ 323.15 K}$$

b. $q'' = h(T_s - T_\infty) = 500(50-20)$

$q = 15000\ W/m^2$

$q'' = \dot{q}(L_A) \Rightarrow \dot{q} = \frac{q''}{L_A} = \frac{15000}{0.3} = \boxed{50000\ W/m^3}$

c. $k_A \frac{d^2T}{dx^2} + \dot{q} = 0$

$\int \frac{d^2T}{dx^2} = \int \frac{-\dot{q}}{k_A}$

$\int \frac{dT}{dx} = \int \frac{-\dot{q}}{k_A} x + C_1$ $\quad [C_1 = 0 \text{ at } x=0]$

$T(x) = \frac{-\dot{q}}{2k_A} + C_2$ $\quad [T(0) = T_w]$

$T(x) = T_w - \frac{\dot{q}}{2k_A} x^2$

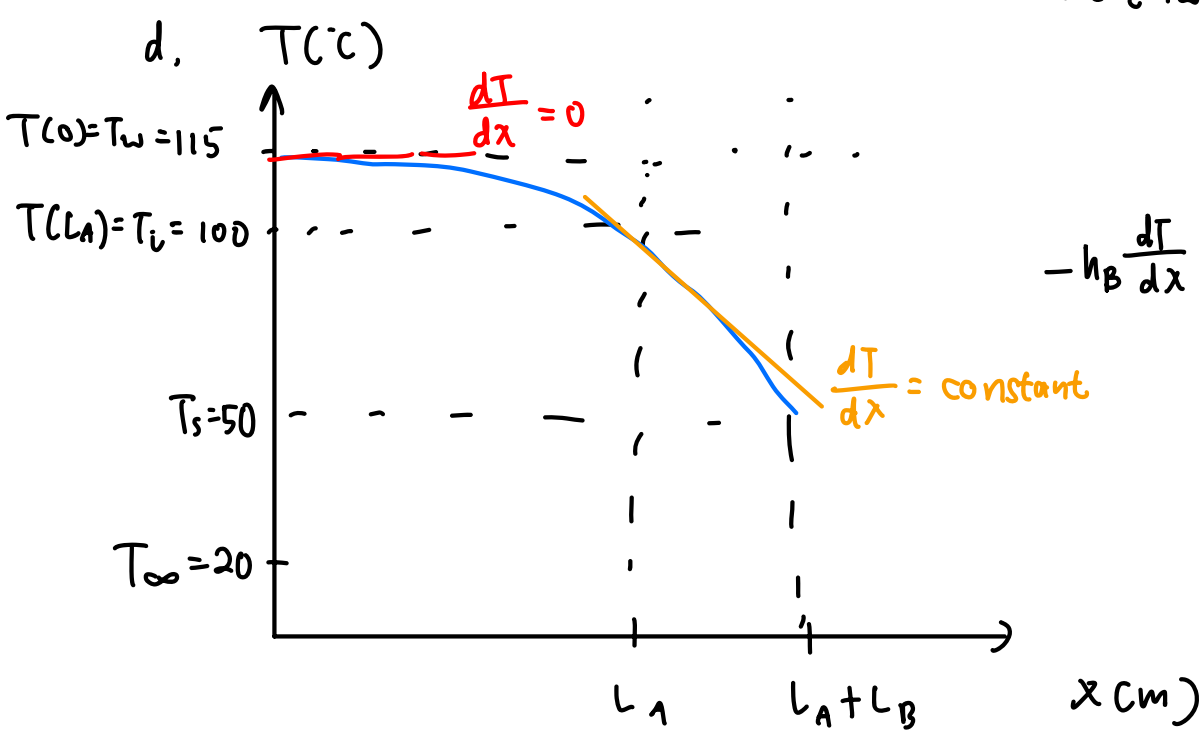
$T(L_A) = T_i$

$T_i = T_w - \frac{\dot{q}}{2k_A}(L_A)^2$

$k_A T_i = k_A T_w - \frac{\dot{q}}{2} L_A^2$

$k_A = \frac{-\dot{q} L_A^2}{2(T_i - T_w)} = \boxed{150\ W/mK}$

d.



$-k_B \frac{dT}{dx} = h(T_s - T_\infty)$

(3) **Given:**

$h_b = 25\ W/mK$ $\quad T_{\infty,i} = 400\ K$ $\quad h_o = 1000\ W/m^2K$ $\quad L_b = 5\ mm$

$k_{TBC} = 1\ W/mK$ $\quad T_{\infty,o} = 1616\ K$ $\quad h_i = 500\ W/m^2K$

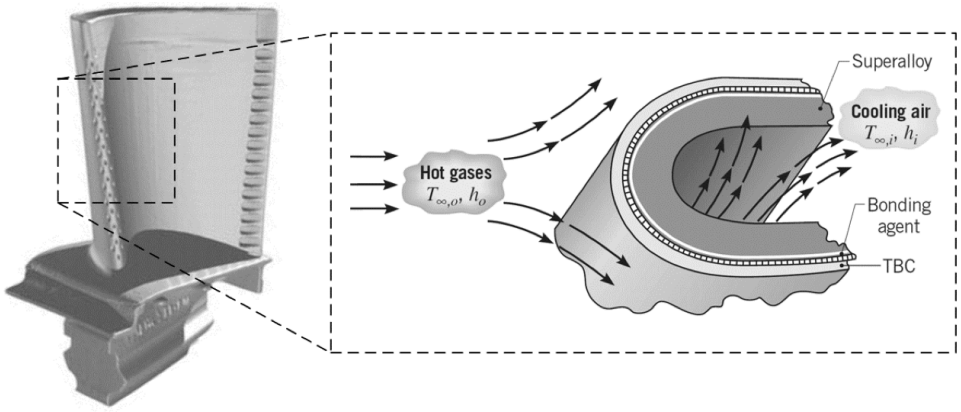
$T_{max} = 1200\ K$ $\quad k'' = 10^{-4}\ \frac{m^2K}{W}$

Find:

a. exterior surface

b. Temp. of superalloy blade

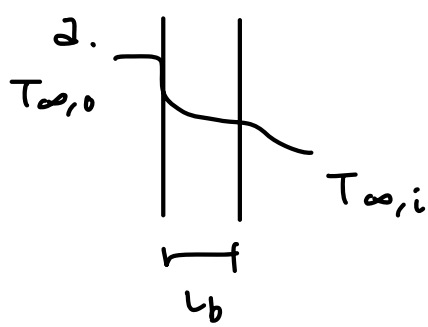
Schematic:



Assumption: 1D heat equation; thermal radiation neglected; turbine blade at steady state, plane wall

Analysis:

a.



$$R''_{tot} = \frac{1}{h_o} + \frac{L_b}{h_b} + \frac{1}{h_i} = \frac{1}{1000} + \frac{0.005}{25} + \frac{1}{500}$$

$$= 0.0032\ m^2K/W$$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{tot}} = 3.8 \times 10^5\ W/m^2$$

$$q'' = h_o(T_{\infty,o} - T_{s,o}) = \boxed{1236\ K}$$

$1236 > 1200K \rightarrow$ can't be maintained below T_{max}

b. $R''_{tot} = \frac{1}{h_o} + \frac{L_{TBC}}{k_{TBC}} + R'' + \frac{L_b}{k_b} + \frac{1}{h_i} = 0.0038\ m^2K/W$

$$q'' = \frac{1616-400}{0.0038} = 3.2\times10^5\ W/m^2$$

$$q'' = \frac{k_{TBC}(T_{s,0}-T_{s,1})}{L_{TBC}} \qquad T_{s,1} = 1296 - \frac{(3.2\times10^5)(0.0005)}{1}$$

$T_{s,1} = \boxed{1136\ K} < 1200\ K$

can be maintained ✓